

mostly characterised by peak vegetation signals in T3 followed by a significant decline in T4. Since this decrease is related to ploughing events or severe frost damage, it appeared rather moderate for frost-resistant winter ICs (Figure 6a). The SAVI-Q50 values of frost-resistant winter ICs thus resembled those of oil-seed rape which also develops a high leaf mass before winter and is marginally affected by frost. However, SAR metrics might resolve this potential source of confusion as they showed distinct VV/VH-MEAN values for oil-seed rape in the last period (Figure 6b). ICs sown late in the year, e.g. after silage maize, developed very little aboveground biomass in T3 with an increasing growth in T4 (Appendix C Figure C1). Consequently, their temporal profiles differed strongly from most winter ICs, making them susceptible to misclassification with winter cereals. Apart from that, the general patterns of winter cereals, fallows and sugar beet clearly differed from that of ICs. Summer ICs, volunteers and weeds were characterised by similar trajectories with high standard deviations across the illustrated metrics. Although summer ICs are typically grown earlier in the year and peak in T2, their average temporal profile was flattened due to the potentially large amount of lucerne and clover samples being cultivated as perennial crops. Weeds and volunteers, on the other hand, are not grown on purpose and can develop very differently on each field. Their individual samples may show a unique pattern or resemble any other reported crop type.

Summarising, the varying development of unmanaged crops (i.e. weeds or volunteers) as well as the incorporation of different main crops – each following a distinct phenology – add to a significant heterogeneity of the target class “other crop types”. Conversely, most IC species and mixtures follow distinct patterns with minor differences depending on their sowing date, frost resistance and growth period. The considered metrics suggest a potential confusion between ICs and oil-seed rape, winter wheat, volunteers and weeds.

Best model and comparison to official statistics

In this section, we focus on the classification of the *Combined* model aggregated to field level which yielded the highest overall and class-wise accuracies. The model was based on STMs from optical and SAR data, as well as on auxiliary information about the cultivated main crop in 2021. ICs were mapped on all field sizes ranging from fine-scale structures to very large parcels, sometimes forming clusters across neighbouring fields. The map displays regional management differences with significantly less IC plots on very fertile soils, e.g. in the Uckermark (Figure 7a). Higher shares of ICs were present in areas with low (Figure 7c) to medium (Figure 7b) soil qualities.

According to the confusion matrix (Table 6), the majority map achieved an OA of 92.9%. The best class-wise accuracies were attained for ICs with both high

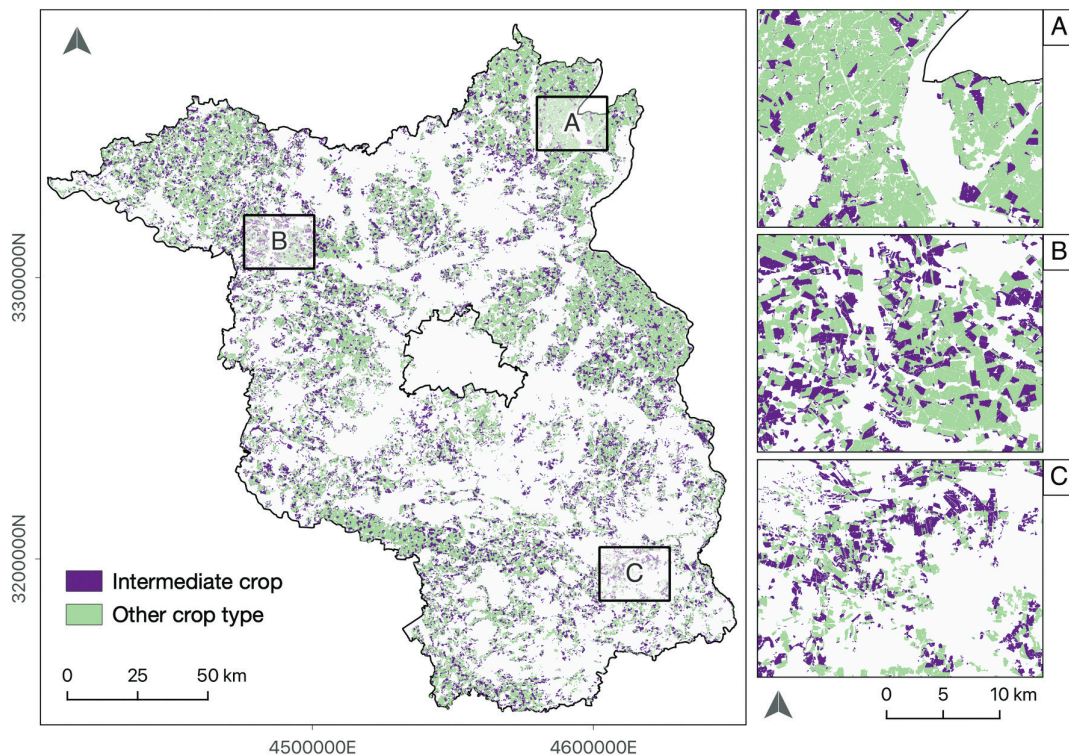


Figure 7. Binary crop type classification of the study area based on the *Combined* model aggregated to field level. The three detailed views are characterised by decreasing soil qualities from a to c.

Table 6. Confusion matrix of the *Combined* classification at field level.

		Reference		Sum
		Intermediate crop	Other crop type	
Prediction	Intermediate crop	153	11	164
	Other crop type	6	69	75
Sum		159	80	239

UA (93.3%) and PA (96.2%). Other crop types were mapped with a comparable UA (92.0%), but lower PA (86.3%). This suggests an overestimation of ICs mainly arising from other crop types being mistaken for ICs. Among the misclassified validation samples, weeds and volunteer grain were most prominent. Additionally, sugar beet and winter cereals introduced confusion with ICs. Vice versa, seed mixtures were partially mapped as other crop types, as were lucerne, white mustard, sunflower and turnip rape.

The overestimation of ICs was also reflected in the obtained area proportion which yielded 32.4% of all cropland. A subsequent exclusion of lucerne and clover fields that were declared as main crops in the IACS reduced the area proportion of ICs to 28.9%. Comparison data from agricultural structure surveys are reported every four to 6 years by the German Federal Statistical Office and were therefore not available for the growing season of interest. Data from previous years reported significantly lower area proportions of ICs with an increasing trend over the last decade (13.2% in 2009/10; 16.2% in 2015/16; 17.9% in 2019/20). Following these records, a realistic area proportion of ICs would amount to approximately 20%. The reported area proportions are statistical estimates based on a sample of 1300 farms in Brandenburg indicating whether an IC was grown between two main crops from June to May of the following year. Both subsidised ICs as well as ICs that were not cultivated in relation to greening measures of the EU were detected.

Discussion

Overcoming challenges of mapping ICs

Despite the challenges of mapping ICs, we were able to capture most phenological differences between ICs and other crop types with the chosen methodology. The derived STMs resembled the anticipated phenological development of crops over time, suggesting that the spectral information was summarised in a meaningful way. Our results confirm the particular importance of autumn and winter months for identifying ICs which has been reported in previous studies using SAR data (Fendrich et al., 2023; Najem et al., 2024) or optical satellite data (Schulz et al., 2021). While optical data densities during winter are critical for IC detection, a combination of data from multiple optical sensors can help overcome this challenge (Lewńska, Frantz, et al., 2024). The higher probability

of missing or low-quality pixels due to clouds during the growing season of ICs additionally justifies the use of STMs. We improved the only existing approach for mapping ICs from optical satellite data (Schulz et al., 2021) we are aware of by implementing more optical-based metrics and by SAR-based input features. Direct comparison to accuracies from other studies is difficult, though, given the wide range of training and validation data used. A consistent definition of ICs across Europe, as proposed by Fendrich et al. (2023), and a publicly available reference dataset are required to improve validation of single studies and comparability between studies. Still, several findings of our study compare well to previous research. In line with Schulz et al. (2021), similar phenological characteristics occur in different IC profiles, indicating that a differentiation of IC species composition is not yet feasible. Moreover, our models encountered problems in detecting weakly developed ICs as found by Najem et al. (2024), especially when grown after silage maize. Beyond that, the use of our own field data enabled the identification of other potential sources of misclassification and allowed for a wide representation of various ICs in the derived map. As a result, our map provides a spatial overview of the cultivation of ICs at field level, no matter if the IC was reported in the IACS data (i.e. registered for subsidies) or not. We hence provide the first comprehensive wall-to-wall mapping of ICs for the study area. The conducted field campaign further allowed to include volunteers and weeds which are neither reported in the IACS nor in official ground-truth campaigns (e.g. Land Use – Land Cover Area Frame Survey (LUCAS)), nor in agricultural structure surveys as used by Fendrich et al. (2023). Unmanaged crops can serve as a bridge for pests and diseases and have a different agronomic and ecological value than ICs. A clear distinction between ICs and volunteers or weeds is therefore necessary, but remains challenging given their considerable spectral-temporal similarities.

Impact of remote sensing-based input features on IC mapping

Based on the results presented, we strongly recommend the incorporation of multiple optical sensors allowing for better data coverage, since optical metrics of the winter periods (T3, T4) were most important for model performance. The primary reason for T3 (October – December) and T4 (January – March)

being most important is that those phases capture critical growth stages of winter ICs and highlight their seasonal variations in contrast to other crop types. Accurate and continuous data during those periods are therefore essential for reliable model performance. NDTI and SAVI were among the most important variables because they provide crucial insights into the vegetation and soil conditions during the winter periods. NDTI is effective in detecting tillage activities and is sensitive to changes in moisture (Quemada et al., 2018), which are key during the post-harvest of main crops and pre-planting phases of ICs. SAVI, on the other hand, adjusts for soil brightness and enhances vegetation signal detection, particularly useful in periods with sparse vegetation cover.

The varying observation density of optical data, however, compromises the robustness of complex distributional metrics (i.e. STD, IQR) across time and space (Lewińska, Ernst, et al., 2024). This questions the reliability of these STMs and might contribute to their low impact on the classifiers. A sensitivity analysis could be performed to better understand the minimum requirements in terms of optical data coverage for mapping ICs. For this purpose, the number of available observations could be artificially reduced step by step while evaluating feature importance and accuracies. Additional use of SAR metrics enhanced classification accuracies as well as robustness and should be favoured over a purely optical classification. Besides reducing misclassification of other crop types, the incorporation of SAR data minimises the effect of varying optical data densities in time and space which may be particularly relevant in years and study areas that yield less CSOs.

Features from T1 (June – Mid-August) yielded low variable importances and SAR metrics from that period were excluded from analysis because they added more confusion. Considering the peak vegetation period could still be valuable for the models since it captures information on the development of the main crops, including harvest, and on early sown ICs. Depending on the availability of optical data in different world regions, it might be useful to modify T1 to either cover a shorter period focusing on maturity and harvest of the main crops (in our case e.g. July – Mid-August), or a longer period covering a major part of the growing season (e.g. April – Mid-August). This adjustment could help to determine the relevance of the first period for future analyses and whether a refined version of T1 is more appropriate.

Limitations

This study encountered several limitations, particularly related to the crop types considered, the overestimation of ICs, and the validation process, all of which are closely tied to the field data collection methodology.

The training data mainly cover winter ICs due to the relatively late start of the conducted field campaign. Each field was only visited once during the campaign which is related to several limitations. First, the field data do not include information on the period over which a crop was grown (e.g. annual or perennial cultivation of lucerne). Second, crops that can be grown as IC based on the timing of tillage (e.g. green rye) were not detected. Third, we may not know if a summer IC was grown before late-sown winter crops, such as winter wheat, were reported. And last, information on the proper development of the crop (e.g. whether farmers decided to break the crop due to winterkill or diseases) was not available. Some of these issues were addressed by closely examining the time series during the data cleaning process, i.e. by excluding field samples that displayed unexpected temporal profiles. However, interpreting individual profiles is highly subjective and does not completely resolve the limitations of the field data.

The overestimation of ICs throughout the study area is partly due to classifying potential perennial species as ICs, particularly in areas with a high percentage of forage crops. On average, the overestimation of ICs could be significantly reduced by excluding lucerne and clover fields from the area assessment. Applying this logic to the map post-classification would reveal that exceptionally high IC shares, as observed in [Figure 7c](#), are unrealistic and result from training perennial crops as ICs. Thus, training perennial crops as other crop types would have been more appropriate and could be achieved by incorporating multitemporal field data that provide additional information (e.g. on cultivation periods, crop development and crop management). Such information would further advance the understanding of other classification errors, especially originating from volunteers and weeds.

Moreover, aggregating the map from the pixel level to the IACS field level assumes spatially consistent management over time (Appendix B cf. Figure B1). If farmers cultivate only parts of a field with ICs, this aggregation leads to misclassification and misrepresentation of the cultivated area. Nevertheless, the accuracy assessment indicates that a greater MMU compensates for mixed pixel effects and intra-field heterogeneity.

Another limitation of the analysis was the lack of independent validation data. For this reason, we split the field data into a training and validation dataset which reduced the number of samples available for building the models. Determining a validation point for each observed crop type was not always possible, and in some cases only a few validation samples were derived. This lack of validation samples substantially limits our ability to assess whether a crop type was correctly assigned to its target class. For example, frost-resistant winter ICs could poorly be validated, with some crop types in this group being completely

misclassified (e.g. turnip rape) and others lacking validation samples (e.g. ryegrass). Additionally, frost-resistant winter ICs showed very similar temporal profiles to oil-seed rape but were trained with fewer samples. Therefore, if a field displays comparable patterns, the models are more likely to predict oil-seed rape. Balancing between crop groups with similar spectral-temporal characteristics, rather than between the two very heterogeneous target classes, could facilitate more plausible predictions and accurate validation.

While our approach does not provide species-specific information, the map provides essential information for modelling soil organic carbon, water and nitrogen carry-over, as well as soil erosion risks. For these purposes, the exact IC species is less important than the overall presence of an IC, as the additional carbon input (soil organic carbon modelling) or the water and nutrient uptake (water and nitrogen carry-over) and degree of soil coverage (soil erosion) varies little among winter ICs. The specific IC species would be important, though, for modelling pest and disease survival (Donatelli et al., 2017), nitrogen fixation by legumes (Liu et al., 2011), and allelopathic effects (An et al., 2003; Dubey & Hussain, 2000; Martins, 2006).

Our study enhances the understanding of patterns from remote sensing for IC mapping and identifies remaining challenges which are important to address in future studies. The IC mapping approach demonstrated here can be applied to agricultural systems similar to our study area. For transferability to other regions, it will be essential to consider local agricultural practices, climatic conditions and satellite data observation densities to make necessary adjustments to the temporal aggregation periods (Pham et al., 2024). The transferability of the approach also relies on adequate ground-truth data for model training and validation, which can be difficult to obtain in data-scarce regions. To address these challenges, future applications could adopt flexible aggregation methods and explore the use of publicly available regional databases or crowd-sourced agricultural data to improve model generalisability.

A significant source of misclassification in our study arose from volunteers and weeds which were often mistaken for ICs. Addressing this issue is critical for improving the reliability of IC mapping and depends on the availability of appropriate training and validation data. The detection of soil tillage prior to the cultivation of ICs, as opposed to volunteers, is one potential solution to enhance discrimination between unmanaged crops and ICs. For this purpose, complete time series or different aggregation periods for SAR data and suitable optical indices, such as NDTI, could improve detection capabilities of soil tillage. Future research should address opportunities to promote the detection of different crop type groups (i.e. summer ICs, winter ICs, frost-resistant winter ICs) or species-specific classification. This could involve exploring grouping strategies for IC types with

similar spectral and temporal characteristics to enhance classification accuracy and reduce misclassification risks. Considering the spectral-temporal similarities of IC species, integration of additional datasets, such as hyperspectral and thermal data might be beneficial (Barnes et al., 2021; Wang et al., 2023). Detailed narrowband hyperspectral data (such as EnMAP or PRISMA) could significantly improve the accuracy of identifying and distinguishing between various plant species. The incorporation of logics based on detailed and multi-annual crop type information (e.g. Blickensdörfer et al. (2022)) could contribute to achieving this objective, e.g. by assuming a higher probability of oil radish in crop rotations with potatoes. Moving forward, it is essential to incorporate additional datasets and test our methods on data from different years and near-real-time scenarios (e.g. Gao et al. (2020, 2023)) to validate the robustness and generalisability of our findings.

Conclusions

Mapping ICs using remote sensing methods presents several challenges related to the diversity of IC species, their spectral and phenological similarity to main crops and varying growing periods. Additionally, the primary growing season for ICs in temperate regions is during autumn and winter, a period marked by frequent cloud cover, limiting the availability and quality of optical remote sensing data.

Despite these challenges, our study demonstrates an effective approach to map ICs by integrating Landsat, Sentinel-2 and Sentinel-1 data. Optical satellite data provided crucial variables for the classification, such as vegetation indices during the key growing stages of ICs. On the other hand, the integration of Sentinel-1 data was particularly important, enhancing the accuracy of the classification by compensating for the limitations of optical data during periods of high cloud cover. Temporal aggregation informed by knowledge of growing seasons emphasised phenological differences between ICs and main crops. Based on a combination of optical metrics, SAR-based metrics and additional information from IACS data, the best classification achieved an overall accuracy of 92.9%, highlighting the value of integrating different remote sensing data sources for IC mapping. The resulting map not only offers valuable spatially explicit information which is crucial for reporting and monitoring sustainable agricultural practices but also provides input for simulation models that generate further information on agricultural production, related emissions and environmental impacts.

Acknowledgments

We gratefully acknowledge the support of Amanda Birkmann and Farina Sempel during the field campaign.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This study contributes to the research projects ModOKlim - Model-based scenario analysis to optimise plant production for climate protection [German Federal Ministry of Food and Agriculture; 2823KLI002] and AdAgriF - Advanced methods of greenhouse gases emission reduction and sequestration in agriculture and forest landscape for climate change mitigation [Ministry of Education, Youth and Sports of the Czech Republic, CZ.02.01.01/00/22_008/0004635].

Data availability statement

The data that support the findings of this study are available from the corresponding author, G.G., upon request.

References

- Amt für Statistik Berlin-Brandenburg. (2021). *Bodennutzung der landwirtschaftlichen Betriebe im Land Brandenburg 2021*. https://download.statistik-berlin-brandenburg.de/998ab5045064dac6/63ca932b2223/SB_C01-02-00_2021j01_BB.pdf
- An, M., Liu, D. L., Johnson, I. R., & Lovett, J. V. (2003). Mathematical modelling of allelopathy: II. The dynamics of allelochemicals from living plants in the environment. *Ecological Modelling*, 161(1), 53–66. [https://doi.org/10.1016/S0304-3800\(02\)00289-2](https://doi.org/10.1016/S0304-3800(02)00289-2)
- Barnes, M. L., Yoder, L., & Khodaei, M. (2021). Detecting Winter cover crops and crop residues in the Midwest US using machine learning classification of thermal and optical imagery. *Remote Sensing*, 13(10), 1998. <https://doi.org/10.3390/rs13101998>
- Bégué, A., Arvor, D., Bellon, B., Betbeder, J., De Aballeyra, D., Ferraz, P. D., Lebourgeois, R., Lelong, V., Simões, M., & Verón, S. R. (2018). Remote sensing and cropping practices: A review. *Remote Sensing*, 10(1), 99. <https://doi.org/10.3390/rs10010099>
- Belgiu, M., & Drăguț, L. (2016). Random forest in remote sensing: A review of applications and future directions. *ISPRS Journal of Photogrammetry & Remote Sensing*, 114, 24–31. <https://doi.org/10.1016/j.isprsjprs.2016.01.011>
- Blanco-Canqui, H., Shaver, T. M., Lindquist, J. L., Shapiro, C. A., Elmore, R. W., Francis, C. A., & Hergert, G. W. (2015). Cover crops and ecosystem services: Insights from studies in temperate soils. *Agronomy Journal*, 107(6), 2449–2474. <https://doi.org/10.2134/agronj15.0086>
- Blickensdörfer, L., Schwieder, M., Pflugmacher, D., Nendel, C., Erasmí, S., & Hostert, P. (2022). Mapping of crop types and crop sequences with combined time series of Sentinel-1, Sentinel-2 and Landsat 8 data for Germany. *Remote Sensing of Environment*, 269, 112831. <https://doi.org/10.1016/j.rse.2021.112831>
- Bowles, T. M., Atallah, S. S., Campbell, E. E., Gaudin, A. C. M., Wieder, W. R., & Grandy, A. S. (2018). Addressing agricultural nitrogen losses in a changing climate. *Nature Sustainability*, 1(8), 399–408. <https://doi.org/10.1038/s41893-018-0106-0>
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324>
- Chen, G., & Weil, R. R. (2010). Penetration of cover crop roots through compacted soils. *Plant and Soil*, 331(1), 31–43. <https://doi.org/10.1007/s11104-009-0223-7>
- Crist, E. P. (1985). A TM tasseled cap equivalent transformation for reflectance factor data. *Remote Sensing of Environment*, 17(3), 301–306. [https://doi.org/10.1016/0034-4257\(85\)90102-6](https://doi.org/10.1016/0034-4257(85)90102-6)
- Dabney, S. M., Delgado, J. A., Meisinger, J. J., Schomberg, H., Liebig, M. A., Kaspar, T., Mitchell, J., & Reeves, W. (2011). Using cover crops and cropping systems for nitrogen management. In J.A. Delgado & R.F. Follett (Eds.), *Advances in nitrogen management for water quality* (pp. 230–281). Soil and Water Conservation Society.
- d'Andrimont, R., Verhegghen, A., Lemoine, G., Kempeneers, P., Meroni, M., & van der Velde, M. (2021). From parcel to continental scale - a first European crop type map based on Sentinel-1 and LUCAS Copernicus in-situ observations. *Remote Sensing of Environment*, 266, 112708. <https://doi.org/10.1016/j.rse.2021.112708>
- Destatis. (2011). *Land- und Forstwirtschaft, Fischerei - Bodennutzung der Betriebe (Struktur der Bodennutzung) - 2010*. https://www.statistischebibliothek.de/mir/servlets/MCRFileNodeServlet/DEHeft_derivate_00004300/2030312107004.pdf
- Destatis. (2017). *Land- und Forstwirtschaft, Fischerei - Bodennutzung der Betriebe (Struktur der Bodennutzung) - 2016*. https://www.statistischebibliothek.de/mir/servlets/MCRFileNodeServlet/DEHeft_derivate_00033419/2030312167004_ergaenz05022018.pdf
- Destatis. (2021). *Land- und Forstwirtschaft, Fischerei - Bodennutzung der Betriebe (Struktur der Bodennutzung) - 2020*. https://www.statistischebibliothek.de/mir/servlets/MCRFileNodeServlet/DEHeft_derivate_00064306/2030212207004.pdf
- DirektZahlDurchfV. (2014). *Verordnung zur Durchführung der Direktzahlungen an inhaber landwirtschaftlicher Betriebe im Rahmen von Stützungsregelungen der Gemeinsamen Agrarpolitik (Direktzahlungen-Durchführungsverordnung - DirektZahlDurchfV) § 31 Flächen mit Zwischenfruchtanbau oder Gründecke (Artikel 46 absatz 2 Buchstabe i der Verordnung (EU) Nr. 1307/2013)*. https://www.gesetze-im-internet.de/dir_ektzahldurchfV/_31.html
- Donatelli, M., Magarey, R. D., Bregaglio, S., Willocquet, L., Whish, J. P. M., & Savary, S. (2017). Modelling the impacts of pests and diseases on agricultural systems. *Agricultural Systems*, 155, 213–224. <https://doi.org/10.1016/j.agry.2017.01.019>
- Dubey, B., & Hussain, J. (2000). A model for the allelopathic effect on two competing species. *Ecological Modelling*, 129(2), 195–207. [https://doi.org/10.1016/S0304-3800\(00\)00228-3](https://doi.org/10.1016/S0304-3800(00)00228-3)
- DWD. (2019). *Klimareport Brandenburg*. Deutscher Wetterdienst. https://www.dwd.de/DE/leistungen/klimareport_bb/klimareport_bb_2019_download.html
- DWD Climate Data Center. (2021). *Phenological observations of crops from sowing to harvest (annual reporters, historical)* (Version v007) [Dataset].
- DWD Climate Data Center. (2022, March 17). *Phenological observations of crops from sowing to harvest (annual reporters, recent)* (Version) [Dataset].
- Eurostat. (2023, August 10). *Glossary: Soil cover*. Statistics Explained. https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Soil_cover
- Faye, B., Webber, H., Gaiser, T., Müller, C., Zhang, Y., Stella, T., Latka, C., Reckling, M., Heckeley, T.,